Apply filters to SQL queries

Project description

In order to investigate potential security vulnerabilities related to login attempts, I will run SQL queries on the database to extract the information necessary for my investigation.

Retrieve after hours failed login attempts

After recently discovering a potential security incident that occurred outside of working hours, I will extract the login attempts by filtering those that failed and occurred after 6:00 PM.

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE login_time > '18:00' AND success = 0;
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  |       0 |
|       20 | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  |       0 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   |       0 |
|       34 | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   |       0 |
|       42 | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   |       0 |
|       52 | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   |       0 |
|       69 | wjaffrey | 2022-05-11 | 19:55:15   | USA     | 192.168.100.17  |       0 |
|       82 | abernard | 2022-05-12 | 23:38:46   | MEX     | 192.168.234.49  |       0 |
|       87 | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 |       0 |
|       96 | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  |       0 |
|      104 | asundara | 2022-05-11 | 18:38:07   | US      | 192.168.96.200  |       0 |
|      107 | bisles   | 2022-05-12 | 20:25:57   | USA     | 192.168.116.187 |       0 |
|      111 | aestrada | 2022-05-10 | 22:00:26   | MEXICO  | 192.168.76.27   |       0 |
|      127 | abellmas | 2022-05-09 | 21:20:51   | CANADA  | 192.168.70.122  |       0 |
|      131 | bisles   | 2022-05-09 | 20:03:55   | US      | 192.168.113.171 |       0 |
|      155 | cgriffin | 2022-05-12 | 22:18:42   | USA     | 192.168.236.176 |       0 |
|      160 | jclark   | 2022-05-10 | 20:49:00   | CANADA  | 192.168.214.49  |       0 |
|      199 | yappiah  | 2022-05-11 | 19:34:48   | MEXICO  | 192.168.44.232  |       0 |
+----------+----------+------------+------------+---------+-----------------+---------+
19 rows in set (0.000 sec)
```

To retrieve this information, I need to use two conditions in my query. I will combine these two conditions with an 'AND' to obtain results that meet both conditions simultaneously.
The '>' operator allows me to extract fields that have a time greater than 18:00.

Retrieve login attempts on specific dates

A suspicious event occurred on 05/09/2022. In order to investigate the incident, I will extract all login attempts that took place on that day and the previous day.

```
MariaDB [organization]> SELECT *  FROM log_in_attempts  WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |       1 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |       1 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |       0 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |       1 |
|       15 | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  |       0 |
|       24 | arusso   | 2022-05-09 | 06:49:39   | MEXICO  | 192.168.171.192 |       1 |
|       25 | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  |       1 |
|       26 | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 |       1 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   |       0 |
|       30 | yappiah  | 2022-05-09 | 03:22:22   | MEX     | 192.168.124.48  |       1 |
|       32 | acook    | 2022-05-09 | 02:52:02   | CANADA  | 192.168.142.239 |       0 |
|       36 | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  |       1 |
|       38 | sbaelish | 2022-05-09 | 14:40:01   | USA     | 192.168.60.42   |       1 |
```

```
|      169 | alevitsk | 2022-05-08 | 08:10:43   | CANADA  | 192.168.210.228 |       0 |
|      170 | sbaelish | 2022-05-09 | 16:43:18   | USA     | 192.168.65.113  |       0 |
|      172 | mabadi   | 2022-05-08 | 08:06:50   | US      | 192.168.180.41  |       1 |
|      178 | sgilmore | 2022-05-08 | 12:27:22   | CAN     | 192.168.52.216  |       0 |
|      184 | alevitsk | 2022-05-08 | 03:09:48   | CAN     | 192.168.33.70   |       0 |
|      186 | bisles   | 2022-05-09 | 04:29:17   | USA     | 192.168.40.72   |       0 |
|      187 | arusso   | 2022-05-09 | 00:36:26   | MEX     | 192.168.77.137  |       0 |
|      189 | nmason   | 2022-05-08 | 05:37:24   | CANADA  | 192.168.168.117 |       1 |
|      190 | jsoto    | 2022-05-09 | 05:09:21   | USA     | 192.168.25.60   |       0 |
|      191 | cjackson | 2022-05-08 | 06:46:07   | CANADA  | 192.168.7.187   |       0 |
|      193 | lrodriqu | 2022-05-08 | 07:11:29   | US      | 192.168.125.240 |       0 |
|      197 | jsoto    | 2022-05-08 | 09:05:09   | US      | 192.168.36.21   |       0 |
+----------+----------+------------+------------+---------+-----------------+---------+
75 rows in set (0.001 sec)
```

To retrieve connection information that matches these two days, I need to use two conditions in my query: one for the date 05/09/2022 and another for 05/08/2022. I will combine these two conditions with an 'OR' to obtain results that match at least one of these two dates.

Retrieve login attempts outside of Mexico

Suspicious login attempts have occurred, but the team has determined that this activity did not originate from Mexico. Therefore, I will extract all login attempts that did not come from Mexico for analysis

```
MariaDB [organization]> SELECT *  FROM log_in_attempts WHERE NOT country LIKE 'MEX%';
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |       1 |
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |       1 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
|        5 | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  |       0 |
|        7 | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 |       1 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |       0 |
|       10 | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 |       0 |
|       11 | sgilmore | 2022-05-11 | 10:16:29   | CANADA  | 192.168.140.81  |       0 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |       1 |
|       13 | mrah     | 2022-05-11 | 09:29:34   | USA     | 192.168.246.135 |       1 |
|       14 | sbaelish | 2022-05-10 | 10:20:18   | US      | 192.168.16.99   |       1 |
|       15 | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  |       0 |
|       16 | mcouliba | 2022-05-11 | 06:44:22   | CAN     | 192.168.172.189 |       1 |
|       17 | pwashing | 2022-05-11 | 02:33:02   | USA     | 192.168.81.89   |       1 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  |       0 |
|       19 | jhill    | 2022-05-12 | 13:09:04   | US      | 192.168.142.245 |       1 |
|       21 | jdunke   | 2022-05-11 | 17:50:00   | US      | 192.168.131.147 |       1 |
|      191 | cjackson | 2022-05-08 | 06:46:07   | CANADA  | 192.168.7.187   |       0 |
|      192 | bisles   | 2022-05-10 | 08:32:03   | USA     | 192.168.201.40  |       1 |
|      193 | lrodriqu | 2022-05-08 | 07:11:29   | US      | 192.168.125.240 |       0 |
|      194 | jclark   | 2022-05-12 | 14:11:04   | CAN     | 192.168.197.247 |       0 |
|      195 | alevitsk | 2022-05-11 | 06:59:13   | CANADA  | 192.168.236.78  |       1 |
|      196 | acook    | 2022-05-10 | 09:56:48   | CAN     | 192.168.52.90   |       0 |
|      197 | jsoto    | 2022-05-08 | 09:05:09   | US      | 192.168.36.21   |       0 |
|      200 | jclark   | 2022-05-12 | 01:11:45   | CANADA  | 192.168.91.103  |       1 |
+----------+----------+------------+------------+---------+-----------------+---------+
144 rows in set (0.001 sec)
```

In order to retrieve this connection information, I need to use the 'LIKE' operator to filter countries whose names start with 'Mex'. I specify that the beginning of the term should be 'MEX' and the rest can vary using the '%' character, resulting in “LIKE 'Mex%’'".

To extract everything that is not from Mexico or starts with 'Mex', I reverse my condition using the 'NOT' term before it.

Retrieve employees in Marketing

The team wants to perform security updates on the computers of some employees in the marketing department. I need to provide them with information about these employees' computers.

```
MariaDB [organization]> SELECT *  FROM employees;
+-------------+--------------+----------+-----------------------+-------------+
| employee_id | device_id    | username | department            | office      |
+-------------+--------------+----------+-----------------------+-------------+
|        1000 | a320b137c219 | elarson  | Marketing             | East-170    |
|        1001 | b239c825d303 | bmoreno  | Marketing             | Central-276 |
|        1002 | c116d593e558 | tshah    | Human Resources       | North-434   |
|        1003 | d394e816f943 | sgilmore | Finance               | South-153   |
|        1004 | e218f877g788 | eraab    | Human Resources       | South-127   |
|        1005 | f551g340h864 | gesparza | Human Resources       | South-366   |
|        1006 | g329h357i597 | alevitsk | Information Technology | East-320    |
|        1007 | h174i497j413 | wjaffrey | Finance               | North-406   |
|        1008 | i858j583k571 | abernard | Finance               | South-170   |
|        1009 | NULL         | lrodriqu | Sales                 | South-134   |
|        1010 | k2421212m542 | jlansky  | Finance               | South-109   |
|        1011 | l748m120n401 | drosas   | Sales                 | South-292   |
|        1012 | m756n668o146 | nmason   | Information Technology | North-160   |
```

```
|        1189 | h784i120j837 | slefkowi | Human Resources       | West-342    |
|        1190 | NULL         | kcarter  | Marketing             | Central-270 |
|        1191 | NULL         | shakimi  | Marketing             | Central-366 |
|        1192 | k5701183m949 | rlaghari | Information Technology | East-138    |
|        1193 | l186m618n319 | esantiag | Information Technology | Central-300 |
|        1194 | m340n287o441 | zwarren  | Human Resources       | West-212    |
|        1195 | n516o853p957 | orainier | Finance               | East-346    |
|        1196 | o225p357q829 | sshah2   | Information Technology | South-385   |
|        1197 | p791q114r509 | aabara   | Information Technology | North-159   |
|        1198 | q308r573s459 | jmartine | Marketing             | South-117   |
|        1199 | r520s571t459 | areyes   | Human Resources       | East-100    |
+-------------+--------------+----------+-----------------------+-------------+
200 rows in set (0.001 sec)
```

First, I start by extracting all the information from the 'employees' table to display the different columns of that table.

```
MariaDB [organization]> SELECT username, department, office FROM employees WHERE department = 'Marketing' AND office LIKE 'East%';
+----------+------------+----------+
| username | department | office   |
+----------+------------+----------+
| elarson  | Marketing  | East-170 |
| jdarosa  | Marketing  | East-195 |
| fbautist | Marketing  | East-267 |
| rgosh    | Marketing  | East-157 |
| randerss | Marketing  | East-460 |
| dellery  | Marketing  | East-417 |
| cwilliam | Marketing  | East-216 |
+----------+------------+----------+
7 rows in set (0.000 sec)
```

Next, I write the above query to retrieve employees in marketing from the East office.

Retrieve employees in Finance or Sales

My team now needs to perform another security update on the computers of employees in the sales and finance departments. For this, I need to provide them with the necessary data.

```
MariaDB [organization]> SELECT username, department, office FROM employees WHERE department = 'Finance' OR department = 'Sales';
+----------+------------+-------------+
| username | department | office      |
+----------+------------+-------------+
| sgilmore | Finance    | South-153   |
| wjaffrey | Finance    | North-406   |
| abernard | Finance    | South-170   |
| lrodriqu | Sales      | South-134   |
| jlansky  | Finance    | South-109   |
| drosas   | Sales      | South-292   |
| jsoto    | Finance    | North-271   |
| jclark   | Finance    | North-188   |
| abellmas | Finance    | North-403   |
| arusso   | Finance    | West-465    |
| iuduike  | Sales      | South-215   |
```

```
| nliu     | Sales      | West-220    |
| sessa    | Finance    | South-207   |
| revens   | Sales      | North-330   |
| sacosta  | Sales      | North-460   |
| bbode    | Finance    | East-351    |
| noshiro  | Finance    | West-252    |
| orainier | Finance    | East-346    |
+----------+------------+-------------+
71 rows in set (0.000 sec)
```

Retrieve all employees not in IT

I now need to provide my team with information about all the computers of employees who are not in the IT department.

```
MariaDB [organization]> SELECT * FROM employees WHERE NOT department = 'Information Technology';
+-------------+--------------+----------+----------------+-------------+
| employee_id | device_id    | username | department     | office      |
+-------------+--------------+----------+----------------+-------------+
|        1000 | a320b137c219 | elarson  | Marketing      | East-170    |
|        1001 | b239c825d303 | bmoreno  | Marketing      | Central-276 |
|        1002 | c116d593e558 | tshah    | Human Resources | North-434   |
|        1003 | d394e816f943 | sgilmore | Finance        | South-153   |
|        1004 | e218f877g788 | eraab    | Human Resources | South-127   |
|        1005 | f551g340h864 | gesparza | Human Resources | South-366   |
|        1007 | h174i497j413 | wjaffrey | Finance        | North-406   |
|        1008 | i858j583k571 | abernard | Finance        | South-170   |
|        1009 | NULL         | lrodriqu | Sales          | South-134   |
|        1010 | k2421212m542 | jlansky  | Finance        | South-109   |
|        1011 | l748m120n401 | drosas   | Sales          | South-292   |
|        1015 | p611q262r945 | jsoto    | Finance        | North-271   |
|        1016 | q793r736s288 | sbaelish | Human Resources | North-229   |
|        1017 | r550s824t230 | jclark   | Finance        | North-188   |
|        1018 | s310t540u653 | abellmas | Finance        | North-403   |
|        1020 | u899v381w363 | arutley  | Marketing      | South-351   |
|        1022 | w237x430y567 | arusso   | Finance        | West-465    |
```

```
|        1183 | b566c710d544 | iquraish | Human Resources | East-400    |
|        1184 | c986d200e170 | ptsosie  | Human Resources | Central-247 |
|        1185 | d790e839f461 | revens   | Sales           | North-330   |
|        1186 | e281f433g404 | sacosta  | Sales           | North-460   |
|        1187 | f963g637h851 | bbode    | Finance         | East-351    |
|        1188 | g164h566i795 | noshiro  | Finance         | West-252    |
|        1189 | h784i120j837 | slefkowi | Human Resources | West-342    |
|        1190 | NULL         | kcarter  | Marketing       | Central-270 |
|        1191 | NULL         | shakimi  | Marketing       | Central-366 |
|        1194 | m340n287o441 | zwarren  | Human Resources | West-212    |
|        1195 | n516o853p957 | orainier | Finance         | East-346    |
|        1198 | q308r573s459 | jmartine | Marketing       | South-117   |
|        1199 | r520s571t459 | areyes   | Human Resources | East-100    |
+-------------+--------------+----------+----------------+-------------+
161 rows in set (0.000 sec)
```

Summary

To assist my team in obtaining the necessary information for analyzing potential security vulnerabilities, I executed these various SQL queries.